The Organization of Environmental Coupling Shapes What Quantum Reservoirs Remember

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For an open quantum reservoir, how the system forgets is part of how it computes. Quantum reservoir computing processes input streams with fixed quantum dynamics and trains only a linear readout. Dissipation can make old inputs fade, but prior studies commonly fix the environmental process and tune only its strength. Here we show numerically that the coupling pattern, meaning whether transitions connect to separate or shared environmental channels, changes which parts of the input history remain accessible. Paired simulations of finite spin reservoirs keep the Hamiltonian, inputs, measurements, and readout fixed. The tested patterns produce distinct task profiles, with no universal winner. Shared relaxation yields the clearest recurring effect, leaving more recent input information accessible than independent local losses. This ordering recurs across system sizes, Hamiltonians, input protocols, and targeted controls. More decisively, accessible memory changes when we alter only the collective combination addressed by an otherwise matched shared channel. Environmental coupling is therefore more than a damping parameter: it is a design layer that shapes not only how quickly information fades, but which input history remains available for computation.

1 Introduction

Temporal computation requires selective memory: recent inputs must remain available long enough to be useful, while distant inputs and the processor's starting state must fade. Reservoir computing achieves this balance with fixed dynamics and trains only a linear map from measured signals to the desired output [1, 2]. Quantum reservoir computing (QRC) uses a quantum system to transform each input into measured features [3]. Because the internal dynamics are not trained, how they retain input history is central to the computation. Purely unitary evolution does not by itself provide the required forgetting. Coupling to an environment can therefore be functional rather than merely detrimental: dissipative dynamics can implement nonlinear transformations whose dependence on older inputs gradually vanishes [4].

Engineered dissipation already prepares and stabilizes states, protects quantum memories, and drives computation [5, 6, 7, 8]. In QRC, noise, damping, optical absorption, measurement back-action, feedback, and reinitialization can create or regulate temporal memory [9, 10, 11, 12, 13, 14, 15, 16, 17, 18]. Studies of coherence and measurements made during a circuit further clarify practically accessible regimes [19, 20]. Sannia *et al.*, for example, showed that tuning uniform, independent qubit relaxation can improve a continuously driven spin reservoir [10]. Related work likewise tunes scalar controls such as drive strength, interaction strength, input duration, and damping rate [21, 22]. These studies establish that dissipation can help, but they largely choose the environmental process in advance and ask how strongly it should act.

An environment specifies more than a rate. It also determines which system transitions lose information separately and which couple through the same channel. With independent relaxation, each qubit has its own route into the environment. With shared relaxation, one channel acts directly on a collective combination of qubit transitions. Other combinations reach it only after the Hamiltonian mixes them together. This distinction is familiar from collective emission and

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structures protected from noise in quantum networks [23, 24]. Its computational consequence for a quantum system driven by inputs is less clear. Does the environment merely set how quickly a processor forgets, or can its pattern of coupling select which parts of an input history remain available?

Here we answer this question numerically for finite reservoirs of driven spins. We show that changing the organization of environmental coupling changes what earlier inputs a fixed readout can recover. We vary which transitions share a channel and how a common, explicitly defined coupling budget is distributed among them. Within each principal pair, the Hamiltonian, input sequence, measured Pauli observables, training procedure, and data split are identical. Across six types of environmental process, no design dominates every benchmark. Instead, different couplings favor memory, nonlinear processing, parity, and chaotic forecasting in different ways. The environment therefore changes the kind of temporal information available to the readout, not simply whether one score improves.

The clearest recurring contrast is between shared and independent relaxation. A shared channel leaves more recent input history accessible to the linear readout and improves recovery of a nonlinear function of that history. Both effects appear in every principal paired instance with five qubits. The ordering remains when the initial settling period, or washout, is extended until the tested dependence on the starting state has vanished to numerical precision. It recurs from four to eight qubits, across four spin Hamiltonian ensembles, and after replacing continuous input driving with encoding through reset operations. It also survives separate comparisons that match average dissipative activity or a representative relaxation timescale, and when each design independently selects its operating point from a bounded range. No obvious scalar measure of dissipation therefore explains the result.

The most direct test changes only the collective transition addressed by a shared channel. Rotating this direction changes accessible memory even though the number and nonzero strengths of the coupled directions, the total weight, and the weight on every qubit all remain fixed. The dependence on coupling direction recurs in a separate study with six qubits. Redistributing coupling weight within one type of process likewise changes memory. Thus even detailed coarse descriptions of dissipation do not fix the computation: which collective transition the environment directly addresses is itself a design variable. The conclusions concern the tested finite reservoirs. They do not establish a universal optimum, an asymptotic scaling law, or a unique microscopic cause. A common coupling budget is a controlled comparison, not complete physical equivalence.

These findings connect QRC with the broader practice of engineering couplings between systems and environments for state preparation, information processing, and memory protection [5, 6, 7, 8, 25, 26, 27, 28]. Here, memory means information about earlier inputs that remains linearly recoverable from the specified Pauli observables after the settling period. It is not a claim about quantum memory lifetime. Within this scope, the architectural implication is clear. The environment is not merely an error source or a clock that sets how quickly information disappears. The transitions it can access shape which parts of the past remain usable. Environmental coupling should therefore be designed alongside the Hamiltonian, input encoding, and measurement. For temporal quantum processing, the environment is not merely the boundary of the processor. It is part of the processor.

2 Reservoir model and controlled environmental coupling

2.1 The fixed driven reservoir

We keep the coherent reservoir fixed and change only its coupling to the environment. The reservoir is the continuously driven network of N interacting qubits introduced in Ref. [10]. During the kth input interval, a scalar value $s_k \in [0,1]$ is held constant for a duration Δt, and the qubits evolve under

$$H(s_k) = \sum_{i<j} J_{ij}\sigma_i^x\sigma_j^x + h\sum_i \sigma_i^z + h(1+s_k)\sum_i \sigma_i^x. \tag{1}$$

The J_{ij} term couples the qubits, the z field sets their static energy scale, and the final term writes s_k into the transverse drive. Each reservoir instance uses one sampled set of couplings J_{ij}. This fixes how the input enters and how information moves before coupling organization is varied.

2.2 Environmental dynamics and reference relaxation

The Hamiltonian describes isolated evolution, while the environment adds a second contribution. For the memoryless model used here, the density matrix obeys the Lindblad master equation [29, 30, 31],

$$\begin{aligned}\dot{\rho} &= \mathcal{L}_s(\rho) = -i[H(s), \rho] + \sum_k \mathcal{D}[L_k](\rho), \\ \mathcal{D}[L](\rho) &= L\rho L^\dagger - \tfrac{1}{2}\{L^\dagger L, \rho\}.\end{aligned} \tag{2}$$

We use Eq. (2) as a target engineered Markovian GKLS generator, not as the generic weak-coupling master equation of an arbitrary common bath. The commutator gives the coherent motion. Each $\mathcal{D}[L_k]$ describes one environmental action on average. The operator L_k identifies the transition and its coefficient sets the strength. Both parts act continuously. Because the dissipative term is quadratic in its jump operator,

$$\mathcal{D}[aL] = |a|^2 \mathcal{D}[L],$$

a coefficient $\sqrt{\gamma}$ makes γ the corresponding rate scale.

The standard model gives each qubit its own relaxation channel. Following Sannia *et al.* [10], we write

$$L_i = \sqrt{\gamma}\, \sigma_i^-, \qquad i = 1, \ldots, N. \tag{3}$$

The lowering operator σ_i^- maps the excited state of qubit i to its ground state. One operator per site gives independent relaxation, while the common $\sqrt{\gamma}$ sets the rate. This is *uniform local relaxation*, the reference point from which we vary coupling organization.

2.3 Generator-level organization of environmental coupling

The organization of environmental coupling has two operational coordinates. A *jump family* specifies which combinations of system operators are directly coupled by the environmental channel. A *rate profile* redistributes the assigned coupling weight within a fixed family.

The distinction between strength and coupling organization is easiest to see with two qubits. Independent local relaxation uses two separate channels,

$$L_1 \propto \sigma_1^-, \qquad L_2 \propto \sigma_2^-,$$

so that the environment acts on the two sites individually. A shared relaxation channel instead uses one combined operator,

$$L_c \propto \sigma_1^- + \sigma_2^-.$$

Here the two relaxation amplitudes are combined before the environmental action occurs. To see the consequence, consider the one-excitation states

$$|+\rangle = \frac{|10\rangle + |01\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|10\rangle - |01\rangle}{\sqrt{2}}.$$

For the symmetric state $|+\rangle$, the two contributions add, and the shared channel directly connects the state to $|00\rangle$. For the antisymmetric state $|-\rangle$, they cancel, so this channel does not relax the state directly. The state is not necessarily protected forever: the Hamiltonian can mix the symmetric and antisymmetric combinations, thereby giving the latter an indirect route to the environment. Nevertheless, the two environmental-coupling designs are physically different. Independent channels expose each site separately, whereas a shared channel exposes only one collective combination directly.

Figure 1 maps these choices across the tested design space. The local–collective example is intuitive, but the distinction must be defined at the generator level.

Let $\{F_\mu\}$ be a fixed set of traceless system operators with

$$\mathrm{Tr}(F_\mu^\dagger F_\nu) = \delta_{\mu\nu}.$$

Expanding every jump operator as

$$L_k = \sum_\mu A_{k\mu} F_\mu, \qquad C = A^\dagger A \succeq 0,$$

makes this distinction invariant. The coefficient $A_{k\mu}$ states how strongly elementary action F_μ contributes to environmental channel k. The matrix C records which elementary actions are coupled and how the coupling weight is arranged among them. Equivalent remixings of the jump list change the individual L_k and A, but leave C, and therefore the dissipative generator, unchanged. Kossakowski geometry here means the support, rank, and orientation of C: a family fixes its operator sector and allowed support, whereas within-family parameters set its nonzero weights and, where applicable, its direction within that support. The construction is invariant under

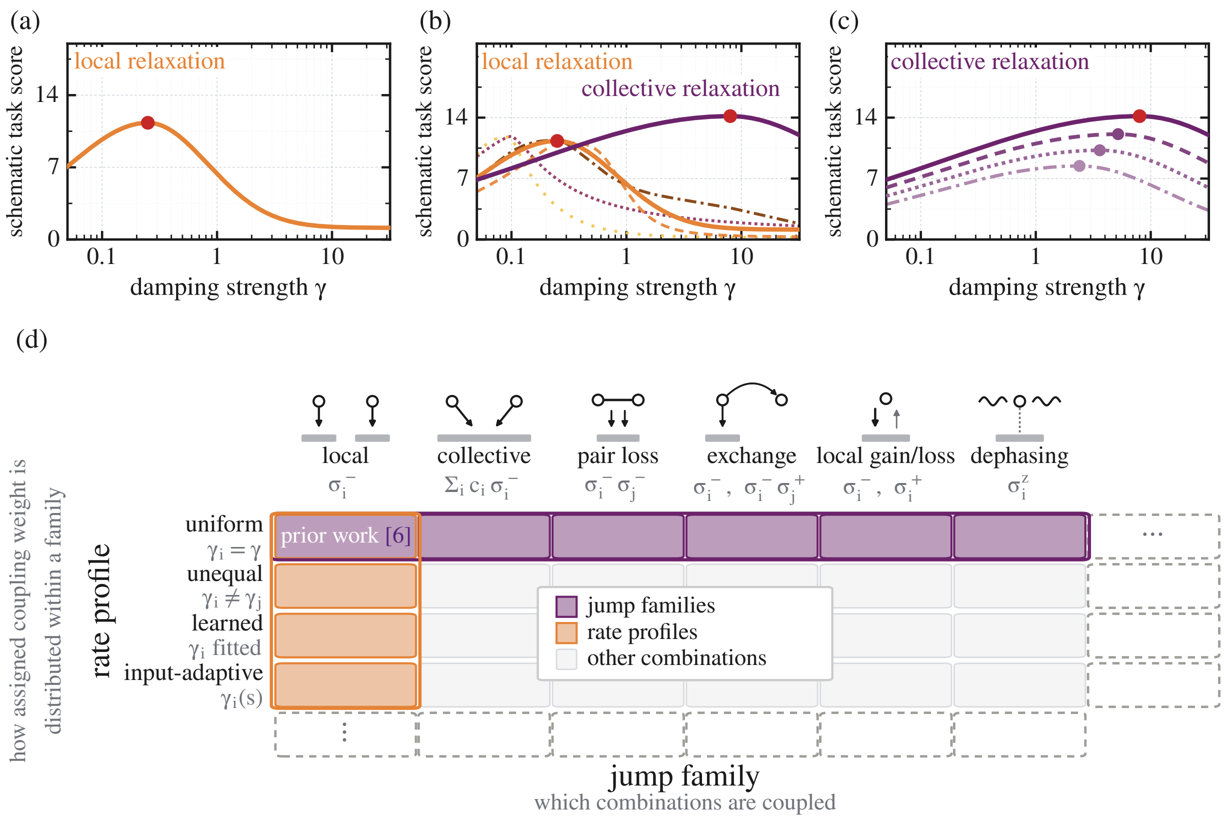

Figure 1: **Environmental coupling has two complementary design coordinates.** *(a)* Previous work primarily varies the overall strength of uniform local relaxation [10]. *(b)* The jump family changes which system combinations couple directly to the environment. *(c)* The rate profile redistributes assigned coupling weight within a selected family. *(d)* Highlighted cells mark the two principal coordinate slices; pale-gray cells show other combinations. Panels (a–c) show schematic task scores, not fitted task data.

equivalent jump-list remixings but remains relative to the declared elementary-operator basis.

For the one-body lowering block, the local–collective distinction takes the form

$$C^{\text{local}}_{\text{lower}} \propto I_N, \qquad C^{\text{collective}}_{\text{lower}} \propto \boldsymbol{c}\boldsymbol{c}^\dagger,$$

where $\boldsymbol{c} = (c_1, \ldots, c_N)^\mathsf{T}$ parameterizes the shared operator

$$L_c \propto \sum_i c_i \sigma_i^-.$$

With nonzero local rates, I_N spans N independently coupled lowering directions, whereas $\boldsymbol{c}\boldsymbol{c}^\dagger$ spans one directly coupled collective direction. Rotating the written local jump operators cannot turn this full-rank block into the rank-one shared block. Thus local and collective relaxation are distinct generators.

Unequal local relaxation instead changes the diagonal weights within the local block and thus moves along the rate-profile axis.

Table 1 is the exact catalogue of the tested environmental actions; Fig. 1 provides their intuitive map. We say gain/loss rather than thermalization because its fixed transitions need not produce a thermal state.

2.4 Common structural normalization and paired comparison

Unlike families contain different numbers and types of jump operators. Assigning the same coefficient to every operator would therefore give more total weight to a family merely because it contains more operators. We instead fix

$$\mathcal{B} = \sum_k \mathrm{Tr}(L_k^\dagger L_k). \tag{4}$$

Each term is the squared operator size assigned to one environmental channel. In the orthonormal basis above, $\mathcal{B} = \mathrm{Tr}\, C$. Fixing $\mathcal{B}$ therefore provides a common structural bookkeeping scale: the total assigned operator weight remains fixed while coupling organization changes.

For the principal fixed-$\mathcal{B}$, common-protocol comparison, we vary only the jump family. The Hamiltonian instance, input sequence, Pauli observables, linear readout, training, selection, and test splits, and assigned operator weight $\mathcal{B}$ remain fixed.

This isolates structural change, not physical equivalence. In the one-body local–collective comparison, fixed $\mathcal{B} = \operatorname{Tr} C$ holds assigned coupling weight fixed while changing the cross-site dissipative correlations encoded in C; it does not equalize activity, the relaxation spectrum, energy flow, entropy production, implementation, or hardware cost. The fixed-$\mathcal{B}$ protocol therefore defines the cross-family structural slice, while activity, gap, and operating-point controls additionally validate only the local–collective memory ordering.

2.5 Readout and computational tasks

After each input interval, local Pauli expectations and same-axis two-qubit correlations summarize the reservoir state,

$$\{\langle X_i\rangle, \langle Y_i\rangle, \langle Z_i\rangle, \langle X_iX_j\rangle, \langle Y_iY_j\rangle, \langle Z_iZ_j\rangle\}.$$

They provide $3N + 3\binom{N}{2}$ features, or 45 at $N = 5$. Fixed-profile jump-family experiments fit only the linear readout; separately identified profile and operating-point studies also select dissipative parameters on data disjoint from final testing. Appendix B gives the ridge control.

Four benchmarks ask what information those features retain. Short-term memory (STM) [32] reconstructs earlier inputs. NARMA-10 is the nonlinear autoregressive moving-average task of order ten. It tests a nonlinear function of recent input history [33]. Parity combines binary memory with nonlinear processing. Mackey–Glass forecasting [34] continues a chaotic signal in closed loop. Capacity is maximized for STM and parity. Error is minimized for NARMA-10 and forecasting.

For the adaptive-profile, finite-sampling, and alternative-control studies, any additional mean matching, estimator choice, calibration, or selection rule is stated explicitly.

Table 1: **Dissipative designs.** The first two rows differ only in rate profile. The displayed jump operators are convenient representatives of the corresponding GKLS generators. Equivalent jump decompositions are not counted as different designs.

design	operators and direct environmental action
uniform local relaxation [10]	$L_i = \sqrt{\gamma}\,\sigma_i^-$, $i = 1, \ldots, N$. The same rate for every qubit; independent sitewise relaxation and the reference design.
unequal local relaxation	$L_i = \sqrt{\gamma_i}\,\sigma_i^-$, $i = 1, \ldots, N$, with the γ_i not all equal. Independent sitewise relaxation with qubit-dependent timescales.
collective relaxation	$\{\sum_i c_i\sigma_i^-\}$. One shared relaxation channel.
pair loss	$\{\sigma_i^-\sigma_j^-\}_{i<j}$. Joint two-excitation removal.
exchange-assisted relaxation	$\{\sigma_i^-\} \cup \{\sigma_i^-\sigma_j^+\}_{i\neq j}$. Local lowering supplemented by incoherent excitation exchange.
local gain/loss	$\{\sigma_i^-, \sigma_i^+\}$. Fixed downward and upward transitions.
dephasing	$\{\sigma_i^z\}$. Phase randomization without excitation loss.

3 Computational consequences of coupling organization

We traverse the two-coordinate design space of Fig. 1: first across jump families, then across rate profiles within a family, and finally across both axes jointly. We then ask whether exact-expectation differences remain visible under finite measurement budgets.

The fixed-weight task map establishes task-dependent computation before we focus on its most strongly replicated consequence: the local–collective contrast in readout-accessible recent-input memory.

3.1 Reference model and how to read the results

Uniform local relaxation is the reference, with one equal-rate lowering channel per qubit. Its family also includes unequal, learned, and input-adaptive profiles. The main comparison uses the equal-phase rank-one collective channel with $c_i = 1$ (Appendix A); profile-optimized models instead learn within-family coefficients.

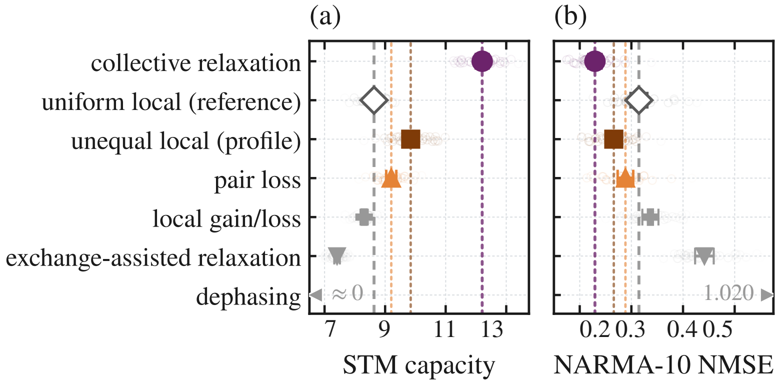

Figure 2: **Different environmental couplings favor different computations.** Scores are shown for continuously driven $N = 5$ reservoirs. Each row represents one dissipative design. Pale circles show individual reservoirs, large symbols show the mean, and horizontal bars show pointwise 95% intervals. The gray diamond and dashed line mark uniform local relaxation as the reference. Colored symbols and matching dotted lines identify every design whose mean outperforms that reference. Higher values are better for STM and parity, whereas lower values are better for NARMA-10 and MG-150. The broken axes in (c,d) enlarge the region where the designs differ while retaining all observations. All designs are compared at the same assigned operator weight $\mathcal{B}$, which provides a common structural reference rather than physical equivalence. Unequal local differs from the reference only through its rate profile.

The four tasks retain their native metrics: STM and parity capacities are maximized, whereas NARMA-10 NMSE and MG-150 MSE are minimized.

Figure 2 reports the absolute $N = 5$ scores in each task's native metric and favorable direction. The dashed vertical line in each panel marks the uniform-local mean; family-colored dotted lines and saturated markers identify all designs with a better mean in that direction. Faint rings show individual reservoir instances. The panels retain their own scales and must be read independently; Appendix Table 2 gives the exact means and standard errors.

3.2 Axis I: Jump-family choice changes the task profile

We first ask whether changing which system combinations couple to the environment alters what the reservoir can compute.

Each paired comparison shares its Hamiltonian, input sequence, operating point, observables, readout, data split, and structural budget $\mathcal{B}$. Only the jump family changes. Uniform local relaxation supplies the reference, while the reset-encoded Fujii–Nakajima (FN) architecture [3] is shown separately as an external reservoir baseline.

Appendix Figure 6 gives the complementary within-task rank view.

The resulting task profiles differ, and no continuously driven design dominates all four tasks. Collective relaxation leads STM and NARMA-10 under the common protocol, but performs worse on parity and long closed-loop forecasting. Exchange-assisted relaxation provides the clearest parity improvement.

Under fixed $\mathcal{B}$, the map shows protocol-specific specialization, not a universal winner. A descriptive STM strength sensitivity preserves the collective lead on the tested grid (Appendix B.8). We focus on the local–collective contrast because it is the cleanest comparison within one lowering action and gives the strongest recurring memory change.

3.3 Main jump-family case study: collective relaxation versus the reference model

We compare complete fixed-budget endpoints with $C_{\text{lower}}^{\text{local}} \propto I_N$ and $C_{\text{lower}}^{\text{collective}} \propto \boldsymbol{c}\boldsymbol{c}^\dagger$; rank, delocalization, coefficient-phase pattern, and drive alignment are not identified as separately causal.

At $N = 5$, collective relaxation increases STM from 8.634 to 12.206. This is an absolute paired gain of 3.57 STM-capacity units and a relative increase of 41%, with a 95% interval of $[3.36, 3.78]$ and 32/32 paired wins. NARMA-10 NMSE decreases from 0.315 to 0.229, corresponding to a 27% reduction in error, and improves in every paired reservoir.

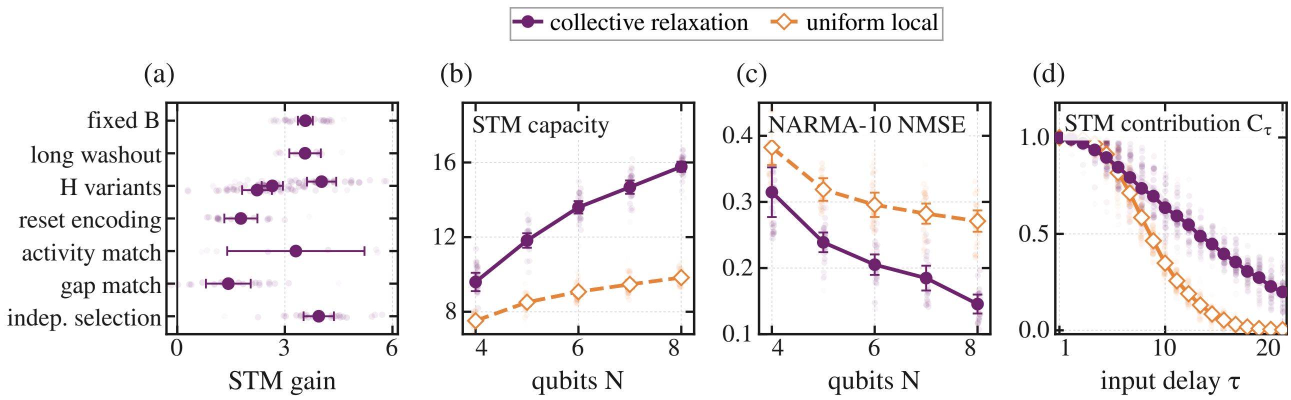

Figure 3: **Collective relaxation retains more usable recent-input memory than the uniform-local reference.** *(a)* Paired STM-capacity gain (collective minus uniform local) under targeted controls; Hamiltonian variants remain separate. *(b,c)* Fixed-initialization STM capacity and NARMA-10 NMSE versus $N = 4, \dots, 8$. *(d)* Delay-resolved STM contribution C_τ. Faint marks and lines show paired or seed-level values; large marks, bars, and bands give means and the declared 95% intervals.

Figure 3(b,c) provides a finite-size recurrence check of the principal $N = 5$ ordering, not an asymptotic scaling analysis. Across 24 paired lineages per size, collective relaxation has the highest mean STM and the lowest NARMA-10 error from $N = 4$ through $N = 8$.

Before interpreting the larger STM score as useful input memory, we test what is being remembered. Removing every jump operator while retaining the Hamiltonian, input protocol, and readout leaves STM and parity near zero, raises NARMA-10 NMSE above one, and preserves initialization dependence (Appendix Table 2). Coherent driving alone therefore reproduces neither the task profile nor controlled forgetting. This unmatched diagnostic contrasts with reset FN, which supplies non-unitary forgetting.

We next distinguish input memory from persistence of the starting state. The same switched input sequence is applied from four widely separated states, and their maximum trace distance is tracked. The convergence records in Appendix B show that local relaxation and pair loss converge across $N = 4, 5, 6$ and remain converged through the 1200-input continuation.

Collective relaxation converges more slowly, but reaches the same regime at the principal $N = 5$ setting and in the tested $N = 6$ lineages. At $N = 5$, it approaches numerical precision by the strict 800-input washout and reaches numerical precision during the specified continuation to 1200 inputs.

A ten-pair subset of the principal $N = 5$ lineages then uses the 800-input washout before STM scoring. Cross-initialization differences are negligible, while collective relaxation retains $+3.564\,[3.124, 4.004]$ STM units with 10/10 wins (Fig. 3(a)). For continuous-drive NARMA-10, all 32 principal pairs are replayed after an independent 600-input prefix, leaving the scored training and test rows unchanged. At the resulting 800-input washout, NMSE is 0.315 locally and 0.230 collectively, a favorable paired difference of $0.0851\,[0.0716, 0.0986]$ with 32/32 wins. The W800-minus-W200 effect change is $-0.00007\,[-0.00057, 0.00043]$. Across four initial states on the first eight pairs, the largest W800 NARMA-10 score spread is 2.18×10^{-5}, far below the smallest paired ground-state improvement, 0.0148. The stable W800 endpoints therefore show that neither ordering is produced by the residual initial-state dependence exposed at W200.

The comparison is also repeated under four spin-Hamiltonian ensembles that change the interaction axis, drive axis, interaction form, or connectivity. Every ensemble retains a positive collective-minus-reference STM difference, ranging from $+2.22$ to $+4.02$ units (Fig. 3(a)). These replications preserve the continuous encoding and Pauli readout, so they establish recurrence within the tested driven-spin architecture. Appendix B describes the Hamiltonian variants and ridge control.

To test whether the ordering is tied to continuous-drive encoding, we replace it with input-by-reset while retaining the $N = 5$ $XX+Z$ processor, Pauli readout, task definitions, and fixed structural budget. After an 800-input washout, collective relaxation increases STM from 4.596 to 6.369, a gain of 1.773 $[1.312, 2.235]$, and decreases NARMA-10 NMSE from 0.673 to 0.491, a favorable reduction of 0.182 $[0.144, 0.220]$, with 16/16 favorable pairs for both endpoints (Fig. 7 and Appendix B.3). The local–collective ordering therefore recurs across two tested input architectures rather than being an artifact of continuous driving.

3.4 Axis II and joint design: rate profiles refine the chosen jump family

Choosing a jump family does not fully specify the environmental coupling. A second choice remains: how should the assigned coupling weight be distributed within that family? We first keep the local-relaxation family fixed and compare uniform, random unequal, learned static, and input-adaptive rate profiles. All sites relax in every candidate, but they need not do so equally. The profile-only unequal-local row in Appendix Table 2 reports the common-protocol comparison; the dedicated STM sweep below uses the separately specified profile-optimization protocol.

The uniform profile assigns the same rate to every site. A random unequal profile introduces a fixed distribution of local timescales. A learned static profile is fitted separately to each reservoir using validation data. An adaptive profile varies with the current input. The static profiles share the same mean rate. The adaptive profile is mean-matched over a fixed input grid but may vary with the input.

Figure 4(a) shows that rate redistribution changes STM without changing the local-relaxation family. Random unequal rates improve on the uniform-local reference, and reservoir-specific learning improves further. The learned-minus-uniform gain is 1.34 units, with an interval of $[1.13, 1.55]$ and 32/32 wins. The adaptive profile gives only a marginal additional gain under the matched restart and evaluation limits. This ordering does not establish a general superiority of static profiles because the static and adaptive parameterizations differ in dimension and stopping behavior.

Within-family tuning redistributes the diagonal rates in the local family, whereas in the collective family it changes the coefficient vector $\boldsymbol{c}$ and thereby rotates the supported rank-one Kossakowski direction; both are well-defined, reproducible tuning operations.

The jump-family and rate-profile axes are not competing alternatives. We therefore cross them directly by comparing uniform and learned profiles within both local and collective relaxation at the same $\mathcal{B}$. Figure 4(b) displays this 2×2 comparison. The vertical separation shows the jump-family effect, the uniform-to-learned slope shows the profile effect, and the different slopes show that the profile benefit depends on the selected family.

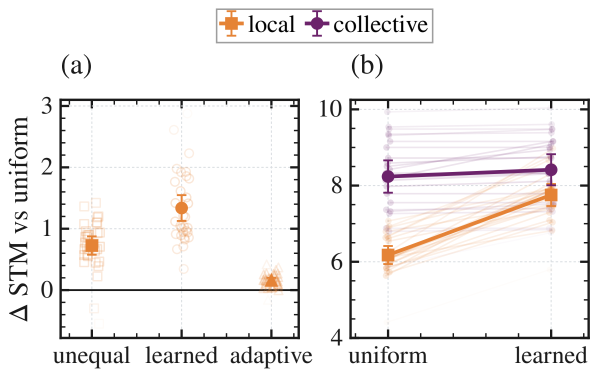

Figure 4: **Rate profiles refine the selected jump family.** *(a)* All 32 paired STM effects for three local rate profiles. Large markers and bars give the mean and 95% interval. *(b)* Paired uniform-to-learned trajectories for local and collective relaxation, showing family-dependent profile gains. Seed marks and trajectories become lighter with distance from their condition means.

The profile gain is larger for local relaxation. The difference between the two gains is 1.397 STM units; all 24/24 paired values are positive, and the familywise 95% interval is $[1.078, 1.716]$. Collective relaxation nevertheless remains higher after both profiles are learned. Profile optimization refines the selected family by a family-dependent amount. Appendix C gives the parameterizations and inference scope.

3.5 Finite sampling determines experimental visibility

The preceding comparisons characterize the computation available to an exact 45-Pauli readout. An experiment observes finite measurement outcomes rather than exact expectation values. We therefore ask a distinct question: at what preparation budget do the differences created by coupling organization become statistically visible?

Finite-sample training, resolvable expressive capacity, and statistical-noise robustness have been studied in QRC and physical learning systems [35, 36]. Our measurement study asks whether the observed ordering survives when expectation values must be estimated from the same nominal number of state preparations. It does not model a platform-specific implementation cost.

At each step, independent sampling spends $N_{\text{prep}} = 45q$ shots over 45 observables; grouped sampling spends the same nominal budget in the global X, Y, and Z bases and reuses each bitstring across 15 features. Each design validates its ridge coefficient before one held-out evaluation. "Preparations per step" excludes replay of the preceding input history, so Fig. 5 studies measurement resolution, not end-to-end runtime.

At the exact-expectation endpoint, Fig. 5 recovers the collective-over-reference memory ordering from the jump-family comparison. Grouped sampling resolves this contrast with one-sixteenth of the preparation count needed by independent sampling. This gain is close to the 15-fold feature reuse built into the estimator. It is a measurement advantage rather than a hardware advantage specific to collective relaxation.

The finite-data curves also show why measurement design belongs in the comparison. Sampling noise does not reduce every STM score by the same amount. At the smallest independent-sampling budgets, pair loss appears best. Unequal local relaxation leads at intermediate precision. Collective relaxation emerges only after the observables are estimated accurately enough. Under simultaneous inference across the complete comparison family, grouped sampling resolves collective relaxation above every competitor only at the largest finite budget. Independent sampling never resolves it as best on the tested grid.

Appendix D gives the estimator definitions, budget parameterization, and simultaneous-inference scope.

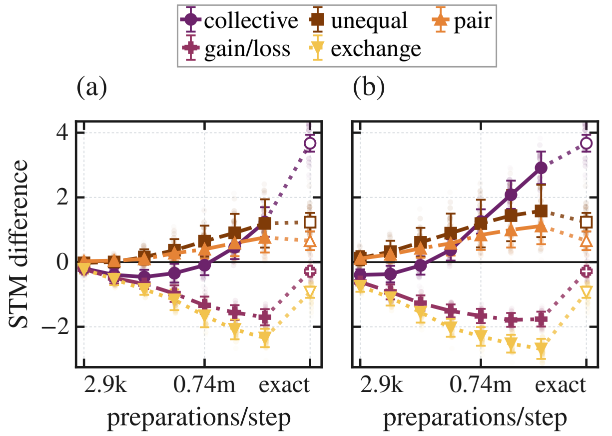

Figure 5: **Finite sampling determines when the jump-family ordering becomes visible.** *(a,b)* Paired STM differences from uniform local relaxation at equal preparation budgets under independent and grouped sampling, respectively. Open markers denote exact readout. Grouping recovers the exact ordering sooner by reusing compatible observables. Faint points show individual paired-seed differences and become lighter with distance from the corresponding mean.

Across these analyses, the memory advantage of collective over uniform local relaxation is the strongest recurring jump-family effect. Section 4 now examines which dynamics are consistent with that contrast.

4 Dynamical interpretation of the local–collective memory contrast

Section 3 establishes the controlled computational ordering independently of the interpretation below. Here we ask which dynamical picture is consistent with it after the tested initial states have been forgotten, not which microscopic ingredient is uniquely causal.

4.1 Working physical picture

The simplest distinction concerns how the reservoir reaches its environment. Under uniform local relaxation, every spin has its own direct relaxation channel. Under collective relaxation, the spins share one channel. The same assigned coupling weight is therefore spread over several separate paths in one case and concentrated into a common path in the other. Some combinations of spin excitations then reach the collective channel only after the Hamiltonian has mixed them with the directly coupled combination. They are not permanently protected: coherent dynamics continue to move information between these combinations. Appendix A gives the corresponding mathematical formulation and states its precise limits. The same interference principle appears in collective emission and in network states that a shared environment cannot address directly [23, 24].

This gives the working picture. A shared environmental channel directly couples fewer combinations; Hamiltonian mixing provides indirect relaxation routes for the others; the resulting slower readout-visible responses can preserve a longer accessible input history.

4.2 Scalar controls for the memory contrast

Fixing $\mathcal{B}$ controls assigned operator weight, but it does not equalize realized jump activity, the relaxation spectrum, or the operating point preferred by each environmental-coupling design. We therefore ask whether any one of these scalar differences is sufficient to explain the principal memory contrast. These controls are not intended to factorize the generator into uniquely causal structural ingredients. Figure 3(a) summarizes the fixed-$\mathcal{B}$, activity-matched, gap-matched, and independently selected comparisons. The STM contrast is positive in all four protocols.

The first control asks whether the shared channel wins simply because it produces a different amount of realized jump activity. Local and collective rates are calibrated separately to the same time-averaged expected activity on an input stream without task labels, then frozen before task scoring. Every calibration passes and the held-out activity difference is unresolved. Collective relaxation still gains 3.306 STM units, with a 95% interval of $[1.393, 5.220]$ and 8/8 paired wins. Different mean jump activity cannot by itself explain the memory contrast. Appendix B.5 gives the calibration protocol.

The second control asks whether the shared channel remembers longer simply because it relaxes more slowly. Before any task stream is generated, each local rate is therefore calibrated to the collective generator's Liouvillian gap at the constant input $s = 0.5$. This gap is the slowest nonzero decay rate of that constant-input evolution and provides one measure of relaxation speed. All 24 matches fall within the prescribed tolerance.

Gap matching reduces the STM and NARMA-10 contrasts by 62.1% and 48.2%, respectively, showing that this timescale explains an important part of the effect. It does not remove the ordering. Collective relaxation retains 1.424 STM units with an interval of $[0.800, 2.049]$ and 20/24 wins, and the NARMA-10 difference remains resolved. Appendix Figure 8(b) resolves the same comparison by input delay. Matching the local rate restores much of its memory tail, but collective relaxation remains higher at long delays. The bands are pointwise 95% intervals across the 24 paired reservoirs. Appendix B.6 gives the calibration protocol.

Finally, a common setting could accidentally favor one jump family. Local and collective relaxation therefore choose their own drive, input duration, dissipative multiplier, and ridge coefficient, with disjoint data for selection and final testing. Collective relaxation still leads by 3.945 STM units, with an interval of $[3.523, 4.368]$ and 24/24 wins. The contrast does not require a shared operating point. The search is bounded rather than global; Appendix B.7 gives the selection protocol and bounded-search scope.

These controls rule out unequal mean activity, one representative decay rate, and a common operating point as sole explanations of the principal local–collective ordering. They do not equalize the complete spectrum, input-dependent propagator products, non-normal transients, energy flow, entropy production, implementation effort, or hardware cost, and they were not repeated for the full cross-family task map.

4.3 Positive dynamical support

Two established ideas guide this interpretation. Fading memory asks whether old inputs and the initial state gradually lose their influence [2, 37, 38]. Information-processing capacity asks which functions of recent input history remain available to a readout [39, 40]. Quantum versions of these ideas have been developed for finite, input-driven, and subsystem reservoirs [41, 42, 43]. Relaxation spectra provide a complementary fixed-input view of how an open quantum system forgets [44, 45]. Because switched inputs mix those responses, we use the spectral calculations as diagnostics rather than as a complete mechanism.

Because STM is obtained by linear regression on the measured feature trajectory, a dynamical component can contribute only if the input excites it and it remains visible in the readout span. Fixed-input mode overlap tests readout visibility; the switched-response kernel additionally includes input excitation. The slower collective modes remain Pauli-visible, and after the representative timescale is matched the switched response is concentrated into fewer observable patterns.

Two additional checks connect this response to the coupling-organization interpretation. Across six process designs, slower midpoint relaxation usually accompanies more STM. A separate sweep mixes local and collective loss continuously; its selected interior point improves on local relaxation in every fresh pair, demonstrating robustness to residual local damping along this path. Together, the diagnostics support one coherent account of the endpoint contrast; Appendix E reports their uncertainty.

4.4 Scope of the explanation

Along the fixed-weight interpolation, every interior one-body lowering block retains Kossakowski rank N while STM changes, so rank alone is insufficient. Learned profiles and the interpolation change several coordinates together. Two matched rank-one interventions isolate Kossakowski orientation. The primary $N = 5$ study finds more STM along the equal-phase end of a prespecified path and against five zero-overlap controls. In a separately generated $N = 6$ replication, rotating the equal-phase vector to the real, sign-balanced direction $(1, 1, 1, -1, -1, -1)$, orthogonal to the uniform site profile of the input drive, lowers STM by 2.271 units $[1.826, 2.717]$, with 24/24 paired wins for equal phase and exact two-sided sign-test $p = 1.19 \times 10^{-7}$. Kossakowski rank, nonzero spectrum and trace, assigned operator weight, sitewise diagonal weights, coefficient magnitudes, and every within-pair processor and data choice remain fixed (Appendix F). These invariants are therefore insufficient to determine memory: Kossakowski orientation relative to the fixed driven processor is an operative coupling-organization coordinate. This does not distinguish alignment with the input from alignment with the coherent dynamics, or make equal phase or a response diagnostic universal.

Dephasing follows a different contraction mechanism. In the connected primary model, it removes the measured signal after washout, which accounts for its near-zero STM. Appendix E gives the driven-dephasing argument. It should not be folded into the shared-channel explanation.

5 Discussion and conclusion

Previous dissipative QRC studies largely fix the environmental process and tune only its strength. We show that this one-rate picture is incomplete. In the finite spin reservoirs studied here, whether qubit transitions couple to separate or shared channels, and how coupling is distributed among them, changes which parts of the input history the measured output can recover. The system–environment connection is therefore part of the reservoir's computational design.

The clearest recurring comparison is between independent and shared relaxation. With independent relaxation, each qubit couples to its own decay channel; with shared relaxation, the environment couples directly to one collective combination of transitions while the Hamiltonian can mix other combinations into it. Shared relaxation makes more recent input history recoverable and improves a nonlinear benchmark that depends on several earlier inputs.

The cleanest intervention changes only the collective combination addressed by an otherwise matched shared channel. Memory changes even though the channels have the same number of coupled directions, the same nonzero coupling strengths, the same total and per-qubit coupling weights, and identical processors, inputs, and readouts. The same direction sensitivity appears at both tested sizes. These comparisons identify what coarse summaries of dissipation miss: the collective transition directly coupled to the environment. Dynamical diagnostics and a gradual path between independent and shared relaxation support this channel-selectivity picture, but do not establish a unique microscopic mechanism.

The local-versus-shared ordering persists after longer washout, across the tested sizes and Hamiltonians, with a different input encoding, and under separate controls for average dissipative activity, a representative relaxation rate, and operating-point selection. These checks make incomplete washout, one processor or encoding, and the matched scalar differences insufficient explanations. They establish a recurring contrast, not a universal advantage over every environmental process.

Across memory, nonlinear processing, parity, and chaotic forecasting, different dissipative designs favor different tasks; redistributing a fixed coupling budget within one design also changes memory. No process is best for every task. Coupling organization is therefore a task-dependent design variable, not a universal optimization rule.

Memory here means previous-input information that is linearly recoverable from the specified Pauli measurements after washout, not an intrinsic quantum-memory lifetime. The full task map concerns numerical five-qubit driven-spin reservoirs. Recurrence from four to eight qubits does not establish asymptotic scaling, and the four-qubit boundary is excluded from claims that the initial state has been fully forgotten. Holding one common measure of coupling weight fixed provides a controlled comparison, not dynamical, physical, or hardware equivalence. The finite-measurement analysis includes projection noise and the chosen estimators, but not drift, gate and readout errors, latency, or full experimental cost. The analytical result on forgetting applies to driven dephasing, not the shared-channel effect, and a generic common bath need not realize the modeled collective process.

The result brings a familiar principle of dissipation engineering, designing the system–environment interaction, into temporal quantum computation [5, 6, 7, 8]. Shared resonators, waveguides, and structured couplers suggest possible routes for testing this channel geometry [25, 26, 27, 28]. The next challenge is to predict useful coupling patterns jointly from the Hamiltonian, input encoding, environment, and measurement, then compare realizable couplers under meaningful physical budgets in larger, noisy reservoirs. How coupling organization interacts with non-Markovian memory is a separate open question.

Together, these results move dissipation in quantum reservoir computing from a parameter to be tuned to a connection to be designed. The question is not only how quickly information should decay, but which transitions should share a route to the environment so that task-relevant history remains readable. For temporal quantum processing, the way a reservoir forgets is part of how it computes.

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A Reservoir model, dissipator construction, and shared methods

This appendix gives the shared definitions and numerical conventions, then the additional controls and interpretation behind each analysis.

A.1 Data and code availability

All code, data, and analysis workflows for this work, together with the manuscript source and figure inputs, are available at https://github.com/eybmits/qrc-dissipation-engineering. The repository documents the full protocol, including random seeds, Hamiltonian and input definitions, data splits, parameter grids, solver and optimization settings, and numerical tolerances. It also contains the calibration records, convergence checks, and per-run outputs behind each reported number, so every result in the paper can be inspected and reproduced.

A.2 Collective process, pairing, and readout

The principal collective process uses one fixed vector in the main, finite-size, Hamiltonian, activity-matched, and gap-matched comparisons:

$$c_i = 1 \quad (i = 1, \ldots, N), \qquad \boldsymbol{c} = (1, \ldots, 1), \qquad L_{\mathrm{c}} = \sqrt{\gamma} \sum_{i=1}^{N} \sigma_i^- .$$

Thus the coefficients are real, have equal phase, and satisfy $\sum_i |c_i|^2 = N$. Only the separately identified profile-by-family check fits a nonuniform collective profile. The common target is the unit-rate local value $\mathcal{B} = N 2^{N-1}$, giving $\mathcal{B} = 80$ at $N = 5$. The finite-size study rematches this target separately at every N.

The jump-family comparison uses independently drawn complete-graph couplings $J_{ij} \sim U[-1, 1]$. Unless a control explicitly changes an operating point, $h = \Delta t = 0.5$, the reservoir starts in $|0 \cdots 0\rangle$, and the STM and NARMA-10 sequences contain 200 washout, 600 training, and 400 test inputs. The task-specific parity and MG-150 protocols are stated below. Comparisons are paired: for a given seed, all dissipative designs see the same Hamiltonian, input sequence, targets, and data split. Variation across seeds therefore reflects both a new reservoir and a new input sequence.

At $N = 5$, the readout contains the expectations of all one-qubit Pauli operators and all same-axis two-qubit Pauli products. These 45 measured features, together with a bias, feed a linear classical readout. The internal quantum dynamics are not trained. Continuous-input evolution is computed by sparse matrix exponentiation. The dense-grid approximation used only for the bounded local-profile search is described in Appendix C.

A.3 Tasks and what they probe

Short-term memory (STM) asks whether a past input can be reconstructed from the present reservoir state. We sum the squared-correlation capacities for delays $1, \ldots, 20$. Larger values mean that more of the recent input history remains linearly accessible.

NARMA-10 adds nonlinear temporal processing. Its target obeys

$$\begin{aligned} y_k = 0.3 y_{k-1} + 0.05 y_{k-1} \sum_{j=1}^{10} y_{k-j} \\ + 1.5 \tilde{s}_{k-10} \tilde{s}_{k-1} + 0.1, \qquad \tilde{s} = 0.2 s, \end{aligned}$$

and is scored by normalized mean-squared error (NMSE), so smaller is better. Parity uses binary input windows of length $1, \ldots, 7$ and 15 virtual nodes per input interval. It has separate 500/250/400 train/validation/test streams after 100 washout inputs. Validation selects the readout regularization before a train-plus-validation refit.

Table 2: **Absolute scores for the fixed-$\mathcal{B}$, common-protocol comparison.** Entries are mean ± standard error. Arrows give the favorable direction, and bold marks the best mean. Row groups distinguish external or diagnostic comparisons, fixed-profile jump-family models, and a profile-only variation within local relaxation. Task definitions and sample sizes are stated immediately above.

model	STM ↑	NARMA-10 ↓	parity ↑	MG-150 ↓
External and diagnostic comparisons				
no dissipation	0.047 ± 0.004	1.308 ± 0.036	0.016 ± 0.001	0.0697 ± 0.0016
reset-encoded FN [3]	10.106 ± 0.260	0.367 ± 0.012	2.295 ± 0.125	0.115 ± 0.008
Fixed-profile jump-family models				
uniform local relaxation [10]	8.634 ± 0.069	0.315 ± 0.008	4.822 ± 0.038	0.0369 ± 0.0029
collective relaxation	$\mathbf{12.206 \pm 0.101}$	$\mathbf{0.229 \pm 0.006}$	3.782 ± 0.028	0.0969 ± 0.0091
pair loss	9.203 ± 0.082	0.289 ± 0.007	4.752 ± 0.037	0.0387 ± 0.0035
local gain/loss	8.307 ± 0.058	0.337 ± 0.008	4.852 ± 0.044	0.0462 ± 0.0038
exchange-assisted relaxation	7.413 ± 0.045	0.441 ± 0.009	$\mathbf{5.083 \pm 0.049}$	0.0518 ± 0.0038
dephasing	0.000 ± 0.000	1.020 ± 0.006	0.000 ± 0.000	0.0599 ± 0.0004
Profile-only variation within local relaxation				
unequal local relaxation	9.845 ± 0.119	0.266 ± 0.007	4.366 ± 0.071	$\mathbf{0.0349 \pm 0.0028}$

The forecasting target obeys the Mackey–Glass equation

$$\dot{x}(t) = \frac{0.2x(t-17)}{1 + x(t-17)^{10}} - 0.1x(t).$$

It is integrated by Euler steps of length one from the seeded delay history $x_t = 1.2 + 0.01\xi_t$, with $\xi_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. Every third Euler iterate is retained, and the first 500 retained samples are discarded. After a 200-sample washout and 600-sample teacher-forced training prefix, the readout is closed for 150 steps. The affine scale is fixed by the washout-plus-training prefix. Only closed-loop predictions fed back to the reservoir are clipped to $[0, 1]$, while the held-out target remains unclipped.

For Table 2, each continuously driven design and the external reset-FN baseline uses 32 reservoirs for STM and NARMA-10, 16 for parity, and 64 for MG-150. The no-dissipation diagnostic uses 30, 30, 32, and 32 reservoirs, respectively.

A.4 Constructing and normalizing dissipative processes

Table 1 gives the physical action of the six process families and the unequal-rate variant. Unequal local rates are drawn log-uniformly from $[0.1, 10]$ and rescaled to unit mean. The collective coefficients are the fixed equal-phase choice $c_i = 1$ stated in Appendix A.2. Pair loss contains every $i < j$. Exchange-assisted relaxation combines local lowering with ordered, excitation-conserving exchange jumps, assigning 40% of $\mathcal{B}$ to the former and 60% to the latter. The local gain/loss process uses downward/upward rates 1/0.3 before the common rescaling. After construction, one scalar rescales every completed design to the same $\mathcal{B} = \sum_k \mathrm{Tr}(L_k^\dagger L_k)$ as unit-rate uniform local relaxation.

Section 2 defines the fixed operator basis and Kossakowski matrix used to compare generators independently of jump-list remixing. For the one-body lowering families, write

$$L_k = \sum_i A_{ki}\sigma_i^-, \qquad \Gamma = A^\dagger A \succeq 0.$$

Then

$$\mathcal{B} = 2^{N-1}\,\mathrm{Tr}\,\Gamma.$$

Thus fixed $\mathcal{B}$ fixes $\mathrm{Tr}\,\Gamma$. Uniform local relaxation has rank N, whereas the equal-phase collective channel has rank one. Kernel directions are Kossakowski coupling directions, not decay modes of the full Liouvillian; Hamiltonian mixing can provide indirect relaxation. The identity above applies only to the one-body lowering block. Across unlike operator sectors, fixed $\mathcal{B}$ remains a structural control rather than an equivalence of spectra, flows, or implementation cost.

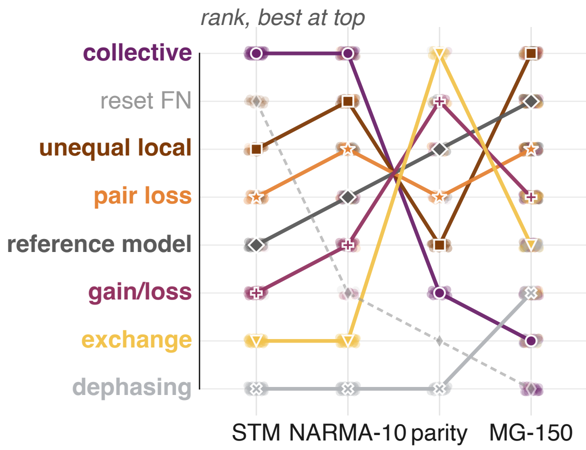

Figure 6: **Task-wise rankings confirm specialization rather than a universal winner among the tested continuously driven designs.** Ranks at $N = 5$ are induced by the absolute means in Table 2; rank 1 is best within each task. Uniform local is the reference. Colors follow the manuscript-wide dissipator palette; lines connect identical designs only as visual guides. Dashed reset-based FN is an external, unpaired baseline. Faint points rank the paired continuously driven seeds against the fixed reset-FN mean; no seed points are assigned to that baseline.

A.5 Statistical reporting

All primary effects are computed as paired, instance-level differences and oriented so that positive means improvement. Unless stated otherwise, we report the mean difference, a two-sided 95% Student-t interval, and the paired win count. For the non-reference cell comparisons, paired sign-flip tests are corrected over the comparison family with the Holm procedure [46]. Finite-size intervals use paired bootstrap resampling. The profile and finite-sampling experiments use simultaneous intervals because many related contrasts are examined together.

B Jump-family computation and local–collective controls

The jump-family analysis asks whether changing the jump family changes computation when the Hamiltonian, input, operating point, readout, and $\mathcal{B}$ are held fixed. The broad map establishes the task dependence of that jump-family choice. The subsequent checks focus on the principal collective-versus-local memory contrast.

B.1 Fixed-$\mathcal{B}$ jump-family task map

Figure 2 and Table 2 report the absolute fixed-$\mathcal{B}$ scores; Figure 6 gives the complementary rank view.

A validation-selected readout control reuses the same STM and NARMA-10 trajectories but divides the 600 fitting rows into 450 training and 150 validation rows. Each design and instance selects ordinary least squares or a ridge value before refitting on all 600 rows. The ordering is unchanged: the collective design remains best on both tasks. Thus the main ranking is not an artifact of imposing one readout penalty on every process.

B.2 Robustness to the coherent dynamics

The principal contrast was repeated in four paired $N = 5$ spin-Hamiltonian ensembles: the primary $XX + Z$ model with an X drive, a $ZZ + X$ model with a Z drive, an $XY + Z$ model with an X drive, and an XX ring with an X drive. The encoding and Pauli readout remain fixed.

B.3 Replication under reset-based input encoding

To test whether the local–collective ordering depends on continuous-drive encoding, we repeat the comparison in a reset-encoded reservoir. At input step t, qubit zero is replaced by

$$\rho_t^+ = |\psi(s_t)\rangle\langle\psi(s_t)|_0 \otimes \mathrm{Tr}_0(\rho_t), \qquad |\psi(s)\rangle = \sqrt{1-s}|0\rangle + \sqrt{s}|1\rangle,$$

after which the autonomous $XX + Z$ Hamiltonian evolves together with either uniform local relaxation or the equal-phase collective channel. This changes the input architecture: the reset is non-unitary and the input-dependent transverse drive is absent during the evolution interval.

The strict comparison otherwise retains the principal $N = 5$ construction: complete-graph couplings independently drawn from $U[-1, 1]$, $h = \Delta t = 0.5$, $\gamma = 1$, $\mathcal{B} = 80$, i.i.d. inputs $s_t \sim U[0, 1]$, the 45 Pauli features and bias, ridge 10^{-8}, STM delays $1, \ldots, 20$, the same NARMA-10 target, and 600 training and 400 untouched test inputs. Within each of 16 fresh pairs, the local and collective reservoirs share the Hamiltonian, input sequence, target, and data split. We use an 800-input washout because collective reset dynamics converge more slowly than their local counterpart. Table 3 reports the strict task endpoints, and Figure 7 shows their paired effects and lag-resolved STM profile.

Table 3: **Strict reset-encoded replication.** The effect is collective-minus-local for STM and local-minus-collective for NARMA-10, so positive values are favorable in both rows. Intervals are paired 95% confidence intervals.

task	local	collective	favorable effect	wins
STM capacity	4.595912	6.369394	1.773 [1.312, 2.235]	16/16
NARMA-10 NMSE	0.672973	0.490933	0.182 [0.144, 0.220]	16/16

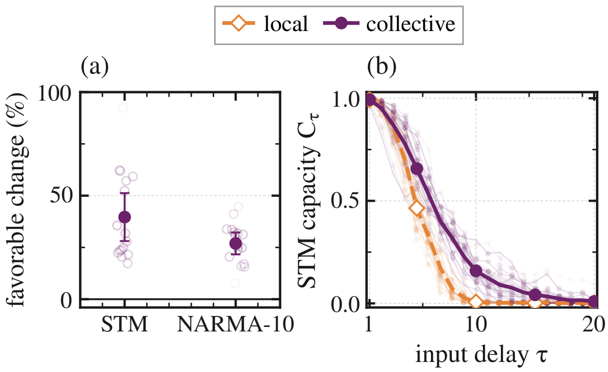

Figure 7: **Replication under reset-based input encoding.** *(a)* Favorable paired relative change with respect to local relaxation for STM capacity and NARMA-10 NMSE across 16 fresh paired reservoirs; positive percentages denote larger STM or lower NMSE. Points show seeds, and filled markers with bars show paired means and pointwise 95% t intervals. *(b)* Lag-resolved STM capacities after the strict 800-input washout; faint lines and points are seeds, and thick lines and bands show means and pointwise 95% t intervals. The fixed-budget comparison uses $\gamma = 1$ and $\mathcal{B} = 80$.

Repeating both tasks and both dissipative models from the ground, fully excited, maximally mixed, and Haar-random pure states gives a maximum score range below 7×10^{-7}; the largest trace distance at the 800-input washout is 1.3×10^{-14}, at numerical precision. The strict ordering therefore reflects retained input history rather than the tested starting state. It establishes that the effect transfers across the two tested input architectures, not universal architecture independence. Nor does this comparison imply that dissipation outperforms a reset-only unitary reservoir; it isolates how two matched environmental-coupling designs shape usable memory.

B.4 Finite size and loss of the initial state

The finite-size study uses 24 fresh paired lineages at every $N = 4, \dots, 8$. Nested principal submatrices of an $N = 8$ coupling draw are rescaled by $\sqrt{4/(N-1)}$, and the Frobenius reference is rematched within each size. The number of measured Pauli features, excluding the fitted bias, grows as

$$F(N) = 3N + 3\binom{N}{2}.$$

The sweep is intended as a finite-size robustness test of the paired local–collective ordering, not as an analysis of asymptotic memory, mixing-time, or Liouvillian scaling. STM is summed over a fixed 20-delay horizon and therefore satisfies $C_{\mathrm{STM}} \le 20$ for every N, whereas $F(N)$ grows with N. Consequently,

$$\frac{C_{\mathrm{STM}}}{F(N)} \le \frac{20}{3N + 3\binom{N}{2}}.$$

Even perfect recall at all tested delays would therefore have a decreasing STM-per-feature ceiling. This ratio is not a neutral scaling or resource-efficiency metric under the present protocol. The grouped readout still requires only the three global X, Y, and Z settings, while the fitted linear readout has $F(N) + 1$ inputs after including the bias. Figure 3(b,c) shows the absolute scores across size.

Because collective coupling can leave some combinations only indirectly damped, we separately tested whether the reported memory could be residual dependence on the starting state. For each process and size, eight fresh lineages at $N = 4, 5, 6$ were driven by identical switched input sequences from the ground, fully excited, maximally mixed, and a Haar-random pure state. Every checkpoint stores the largest trace and Pauli-feature distances over the six initial-state pairs. Local and pair processes converge rapidly. A continuation follows all 48 local and pair lineages at $N = 4, 5, 6$ to 1200 inputs; their worst trace distance over steps 1100–1200 is 6.85×10^{-15}.

For collective relaxation at the principal $N = 5$ setting, the worst trace distance falls from 4.78×10^{-4} after 200 inputs to 1.32×10^{-11} after 800. A separate continuation follows all eight lineages to 1200 inputs and reaches 4.42×10^{-15}, below its declared 10^{-14} gate. The slowly forgetting collective $N = 4$ boundary case is excluded from initialization-independent claims.

A stricter STM check uses ten principal-pair lineages, three distant initial states, and washouts of 50, 200, and 800 inputs. A separate continuous-drive NARMA-10 check uses all 32 principal pairs. For its W800 condition, a frozen 600-input prefix is placed before the unchanged canonical 1200-input sequence; the final 200 washout inputs and the 600 training and 400 test rows are thus identical to W200. Local and collective reservoirs share every input, target, split, Hamiltonian, and readout choice within a pair. Ground-state scoring uses all 32 pairs, and the first eight repeat both washouts from the ground, fully excited, maximally mixed, and Haar-random states. The W200 scores must reproduce the sealed principal values before the W800 aggregate is accepted. At W800, local and collective NMSE are 0.315 and 0.230, respectively; the local-minus-collective difference is 0.0851 $[0.0716, 0.0986]$, favorable in all 32 pairs. Its change from W200 is $-0.00007\,[-0.00057, 0.00043]$. The largest four-state collective score spread at W800 is 2.18×10^{-5}, below the smallest paired ground-state improvement, 0.0148. The archive records every checkpoint and the serialization-only amendment made before any non-audit score was generated.

B.5 Matched expected jump activity

The fixed-$\mathcal{B}$ comparison deliberately allows the realized jump activity to differ. To test whether this alone explains the memory result, we calibrate the two processes using the time-averaged expectation

$$\mathcal{J} = \frac{1}{T\Delta t} \sum_{t=1}^{T} \int_0^{\Delta t} \sum_k \mathrm{Tr}\left[L_k^\dagger L_k \rho_t(\tau)\right] \mathrm{d}\tau.$$

Local and collective rates are fitted separately to $\mathcal{J}_\star = 0.5075$ on a shared, unlabeled input stream that is independent of every task stream. After 100 washout inputs, activity is averaged over 150 intervals. All 16 calibrations for eight fresh reservoir pairs pass the predeclared 1.5% tolerance before task scoring.

B.6 Matched midpoint Liouvillian gap

A second control asks whether the result is only a consequence of the slowest relaxation time at a representative constant input. For each of 24 fresh $N = 5$ Hamiltonians, the local rate is calibrated to the collective Liouvillian gap at $s = 0.5$ before task inputs are generated. All calibrations satisfy the 0.5% relative-error gate.

B.7 Independent operating-point selection

The third control lets local and collective relaxation select their own operating points without using test data. The bounded search varies the drive, input duration, dissipative strength, and readout penalty over predeclared grids. A two-reservoir prescreen advances eight settings per jump family. Twelve disjoint reservoirs then select one setting and readout penalty per process. The final 24 reservoirs are opened only after these choices are written. The collective strength choice is additionally bracketed on the selection set before the final test.

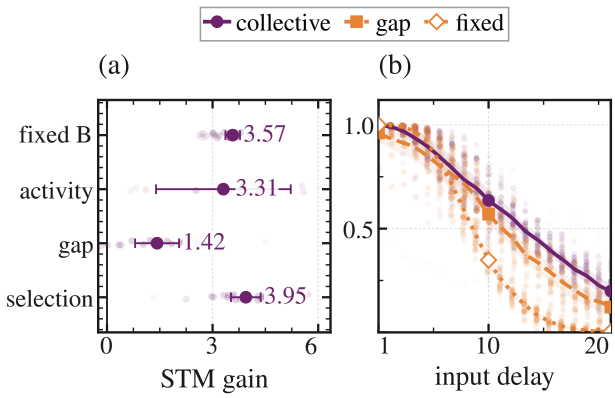

Figure 8: **Targeted scalar controls preserve the local–collective memory contrast.** *(a)* Collective-minus-reference STM effects for the fixed-$\mathcal{B}$, activity-matched, gap-matched, and independently selected protocols. *(b)* Lag-resolved STM at fixed $\mathcal{B}$ and after matching the local rate to the collective midpoint Liouvillian gap. Faint points show paired seeds; bars and bands show the declared 95% intervals.

B.8 Validation-selected scalar-strength sensitivity

The fixed-$\mathcal{B}$ family map is a controlled jump-family slice rather than a globally optimized ranking. In a complementary post hoc STM sensitivity analysis, scalar strength is selected separately for six non-dephasing process designs by leave-one-reservoir-out validation. All designs use the base multiplier grid $\{0.05, 0.1, 0.25, 0.5, 1, 2, 4\}$; uniform local and collective relaxation also include 8 and 16, which bracket the collective optimum at 8. The selected multipliers are 8 for collective, 0.1 for local gain/loss and exchange-assisted relaxation, and 0.25 for uniform local, unequal local, and pair loss; every curve is bracketed. Collective relaxation has the largest mean across the 20 held-out scores, and 50,000 selection-aware bootstrap draws give Bonferroni-simultaneous 95% percentile intervals whose lower endpoints are positive for its advantage over every competitor. This supports the principal STM lead across the tested strength grid, but it does not constitute a fully disjoint, task-wise optimized family ranking or a four-task Pareto analysis.

C Rate-profile optimization and comparison

The rate-profile analysis keeps the jump family fixed and changes only how local relaxation strength is distributed across sites. It therefore tests the second coordinate of environmental-coupling organization independently of the jump-family comparison.

C.1 Within-family profile sweep

The profile sweep uses $N = 5$, mean local rate 1.5, $(h, \Delta t) = (0.5, 0.5)$, and a shorter protocol tailored to profile optimization. It uses 100 washout inputs, 100 optimization-training inputs, 150 validation inputs, and 200 test inputs. All static profiles have the same arithmetic mean. We compare a uniform reference with a random unequal profile and a static profile learned on validation data. A fourth model uses the input-adaptive profile $\gamma_i(s) = \mathrm{softplus}(a_i s + b_i)$. The adaptive profile is matched only in mean over the frozen input grid, so its instantaneous Frobenius strength may vary.

Static and adaptive profiles are optimized from seeded starts with fixed evaluation limits and stopping rules.

C.2 Does profile learning act the same way in every family?

A separate 2×2 check crosses local versus collective relaxation with uniform versus learned profiles. Figure 4(b) displays the 24 paired trajectories and their mean changes. Every candidate is normalized to the same $\mathcal{B}$, and exact continuous-input evolution replaces the dense approximation. For local relaxation, learning redistributes diagonal sitewise rates. For collective relaxation, it changes the coefficient vector $\boldsymbol{c}$ and thereby moves the supported one-dimensional Kossakowski direction while preserving rank one. The axes are therefore operational and interacting design coordinates rather than a globally orthogonal factorization. Inference uses simultaneous coverage over the eleven declared contrasts.

D Finite-sampling estimators and inference

The finite-sampling analysis asks whether the exact-expectation ordering remains visible when Pauli features must be estimated from a finite number of preparations. It compares two estimators at the same nominal preparation count, rather than treating shot noise as independent additive noise on every feature.

D.1 Two estimators at equal preparation count

At $N = 5$, write $N_{\mathrm{prep}} = 45q$. The independent estimator spends q shots on each of the 45 observables. The grouped estimator instead spends $15q$ shots in each global X, Y, and Z product basis and reuses each measured bit string for all compatible one- and two-qubit features. Table 4 emphasizes the methodological difference without repeating the budget grid.

Table 4: **Finite-sampling estimators at equal nominal preparation count.** Grouping changes how measurement records are reused and therefore preserves correlations between compatible Pauli estimates.

	independent observables	grouped Pauli settings
recorded data	one binomial estimate per feature	full bit strings in X, Y, and Z bases
record reuse	none across features	one record informs all compatible features
covariance	omitted by construction	retained within each setting
modeled scope	projection noise	projection noise with grouping, without hardware noise

D.2 Training and simultaneous inference

Six non-dephasing designs are evaluated on 20 paired reservoir and measurement instances under both estimators. Validation selects the readout penalty from the predeclared grid before a train-plus-validation refit. We test $q = 64 \times 4^j$ for $j = 0, \ldots, 6$, so both estimators receive $45q$ preparations per step. Finite-budget intervals are simultaneous over every declared design-pair, estimator, and budget contrast. The exact-expectation endpoint is displayed separately.

The curves in Figure 5(a,b) should therefore be read as a resolution study. A design is distinguished from local relaxation only when the simultaneous interval excludes zero, not merely when its mean is larger. The model does not include acquisition latency, device drift, or gate and readout errors.

E Dynamical interpretation: support and limits

The following diagnostics test whether the observed response is consistent with the shared-channel interpretation without claiming a unique microscopic cause.

E.1 Slow dynamics that remain visible to the readout

For eight $N = 5$ local–collective lineages, the slowest right modes at five constant inputs are projected onto the 45-Pauli readout span. Collective relaxation retains slower modes with larger overlap. The paired differences are $0.338\,[0.239, 0.436]$ in decay rate and $0.104\,[0.028, 0.180]$ in readout-weighted retention. Because this omits excitation weights and switched propagator products, it supports but does not uniquely identify the shared-channel interpretation.

E.2 Concentrated response under switched input

Starting from synchronized trajectories, a complementary diagnostic applies small input perturbations and follows the 45-feature response over 20 lags. Collective relaxation concentrates this response in fewer observable patterns than the matched local process. Its effective rank is lower by $1.270\,[0.959, 1.580]$, while the leading-pattern energy fraction is higher by $0.285\,[0.243, 0.327]$. The switched-input result supports the same picture. It is an empirical response kernel rather than a complete modal decomposition.

E.3 A gradual path from local to collective coupling

To determine whether the endpoint contrast emerges along a continuous coupling-organization path, we interpolate the dissipators at fixed Frobenius weight as $\mathcal{D}_\alpha = (1-\alpha)\mathcal{D}_{\text{local}} + \alpha\mathcal{D}_{\text{collective}}$, with $0 \le \alpha \le 1$. Interior points are ranked on separate gap-only reservoirs before fresh task streams are opened. Increasing the collective component slows midpoint relaxation and raises STM at $N = 4$ and 5. The selected interior point beats local relaxation in every fresh pair. This links the endpoint result to a continuous within-family deformation, not to a general selector for unrelated processes.

E.4 Why driven dephasing forgets uniformly

The near-zero STM of the dephasing control admits a direct contraction argument under the connectivity assumptions of the primary model.

Proposition 1 (Uniform contraction of the driven dephasing model). *Let $L_i = \sqrt{\gamma_i} Z_i$ with every $\gamma_i > 0$, let $\Phi_s = \exp(\mathcal{L}_s \Delta t)$ with $\Delta t > 0$, and suppose the nonzero single-spin-flip matrix elements of $H(s)$ connect the computational-basis hypercube for every $s \in [0,1]$. Then there is a $\kappa < 1$ such that*

$$\sup_{s\in[0,1]} \left\| \Phi_s|_{\mathrm{Tr}=0} \right\|_{2\to 2} \le \kappa.$$

Here $\|X\|_2 = (\operatorname{Tr} X^\dagger X)^{1/2}$. *Hence* $\|\rho_n - I/2^N\|_2 \le \kappa^n \|\rho_0 - I/2^N\|_2$ *for any switched input sequence and initial state.*

Proof. For $X_t = \exp(t\mathcal{L}_s)X$, the Hamiltonian commutator is skew-adjoint in the Hilbert–Schmidt inner product, while the Hermitian-unitary jumps give $\frac{\mathrm{d}}{\mathrm{d}t}\|X_t\|_2^2 = -\sum_i \gamma_i \|[Z_i, X_t]\|_2^2 \le 0$. Equality at Δt would make the norm constant, so every $[Z_i, X_t]$ vanishes and X_t remains diagonal. Its derivative can remain diagonal only if $[H(s), X_t] = 0$. Hypercube connectivity then makes X_t scalar. Every nonzero traceless operator therefore contracts strictly. Attainment in finite dimension, continuity in s, and compactness of $[0, 1]$ give one common $\kappa < 1$. □

State differences are traceless, so the bound applies to every switched sequence and all $s \in [0, 1]$. It explains the dephasing zeros in Table 2 for the connected primary model, not arbitrary unital dissipators.

F Orientation within matched rank-one channels

To vary the supported collective direction at fixed coarse channel invariants, we use

$$c_i(f) = \exp[2\pi \mathrm{i} f(i-1)/5]\,, \qquad f \in \{0, 0.25, 0.5, 0.75, 1\}.$$

Here $f = 0$ is equal phase and $f = 1$ is orthogonal to the uniform direction. Every condition has Kossakowski rank one, spectrum $(5, 0, 0, 0, 0)$, unit diagonal, $|c_i| = 1$, magnitude IPR $1/5$, and $\mathcal{B} = 80$; four frozen phase-scrambled controls add zero-overlap directions.

The confirmation uses 32 fresh paired reservoirs, excludes the pilot, and shares Hamiltonian, input, target, split, and readout across all nine conditions within each pair. After an 800-input washout, the unchanged protocol uses 600 training and 400 untouched test inputs, delays 1–20, 45 Pauli features, and fixed ridge 10^{-8}. Four-state audits pass on the first eight lineages.

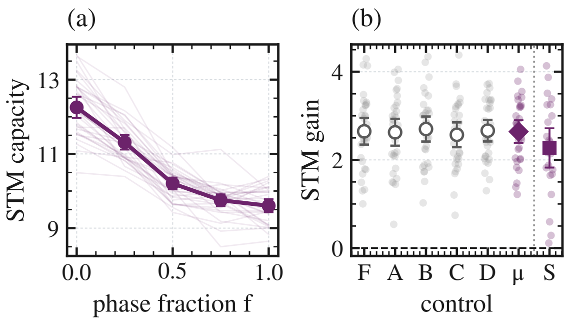

Figure 9: **Accessible memory depends on the matched rank-one direction.** *(a)* Prespecified $N = 5$ phase path. *(b)* Equal-phase-minus-control effects for five zero-overlap directions, their within-pair mean μ, and the $N = 6$ sign-balanced replication S. Marks show pairs, means, and 95% t intervals (32 and 24 pairs).

Equal-phase STM is 12.253, versus 9.604 at the Fourier endpoint, a paired difference of 2.648 [2.347, 2.950] with 32/32 wins and sign-flip $p = 1.0 \times 10^{-5}$. Against the within-pair mean of all five zero-overlap directions, the gated effect is 2.642 [2.383, 2.900], again with 32/32 wins and $p = 1.0 \times 10^{-5}$. The ordered path and validation-selected-ridge sensitivity agree.

Real sign-balanced replication at a second finite size. A separate 24-pair $N = 6$ endpoint replication compares $c_+ = (1, 1, 1, 1, 1, 1)$ with $c_\perp = (1, 1, 1, -1, -1, -1)$, where $c_+^\dagger c_\perp = 0$. At $\mathcal{B} = 192$, jump count, rank-one spectrum $\{6, 0, 0, 0, 0, 0\}$, trace, diagonal weights, coefficient magnitudes, processor, input, readout, data, and ridge rule remain fixed. All 24 ground/mixed-state audits and the first six four-state audits pass at W800. Equal-phase STM is 13.626 versus 11.354, a paired gain of 2.271 [1.826, 2.717] with 24/24 wins and exact sign-test $p = 1.19 \times 10^{-7}$; every delay-wise mean difference is positive. Thus orientation affects accessible memory beyond these coarse matches, without implying scaling, global optimality, or a unique mechanism.